THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 7 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 8: **Working with Fractions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. $\frac{3}{4} \div (\frac{3}{2} \div \frac{1}{2})$ equals: 1

(a) $\frac{1}{4}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{1}{8}$ **(d)** 1

2. $\frac{2}{5}$ of a number is 18. Three times the number is: 1

(a) 135 **(b)** 270 **(c)** $135\frac{1}{2}$ **(d)** $67\frac{1}{2}$

3. A pipe fills $\frac{2}{5}$ of a tank in 12 minutes. The time to fill the whole tank (minutes) is: 1

(a) 15 **(b)** $30\frac{1}{2}$ **(c)** 30 **(d)** 60

4. $\frac{1}{3}$ of a wall is already painted. Then $\frac{1}{2}$ of the remaining part is painted. The fraction of the whole wall still unpainted is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{7}{12}$

5. If $0 < x < 1$, then $\frac{1}{x}$ is: 1

(a) negative **(b)** greater than 1 **(c)** equal to 1 **(d)** less than 1

6. $\frac{3}{5}$ of $3\frac{1}{3}$ is: 1

(a) 2 **(b)** $3\frac{14}{15}$ **(c)** 4 **(d)** 1

7. If $x \times \frac{3}{4} = \frac{9}{16}$, then $x \div \frac{3}{4}$ is: 1

(a) $\frac{1}{2}$ **(b)** 1 **(c)** $1\frac{1}{2}$ **(d)** 2

8. $\frac{5}{6} \div (\frac{5}{3} \div \frac{2}{3})$ equals: 1

(a) $\frac{3}{4}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{1}{3}$

9. **Assertion (A):** $\frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$. 1

Reason (R): The product of two fractions is (product of numerators) / (product of denominators).

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** $\frac{3}{4} \div \frac{1}{2} = \frac{3}{2}$. 1

Reason (R): $\frac{1}{2}$ of $\frac{3}{4}$ is $\frac{3}{8}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Simplify $\left(\frac{5}{6}+\frac{1}{3}\right) \div \left(\frac{5}{6}-\frac{1}{3}\right) \times \frac{1}{2}$. 2

12. After $\frac{2}{7}$ of a number is removed, 35 is left. Find the number and the part removed. 2

13. Find the fraction exactly halfway between $\frac{1}{3}$ and $\frac{1}{2}$, and verify that its distance from each is the same. 2

14. Simplify $\left(\frac{7}{8}+\frac{3}{8}\right) \div \left(\frac{7}{8}-\frac{3}{8}\right) \times \frac{4}{5}$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Ravi alone can finish a job in 6 days and Sita alone in 3 days. What fraction of the job do they finish together in one day, and how many days do they take together? 3

16. A cyclist rides at $\frac{9}{2}$ km per hour for 2 hours, then at $\frac{15}{4}$ km per hour for 4 hours. Find the total distance and the average speed for the whole journey. 3

17. A pole of 20 m is cut into pieces of $1\frac{2}{3}$ m each. How many pieces are obtained and how much is left? 3

OR

Ravi alone can finish a job in 10 days and Sita alone in 15 days. What fraction of the job do they finish together in one day, and how many days do they take together? 3

18. A cyclist rides at $\frac{21}{2}$ km per hour for 4 hours, then at $\frac{35}{4}$ km per hour for 2 hours. Find the total distance and the average speed for the whole journey. 3

OR

A pole of 25 m is cut into pieces of $2\frac{1}{4}$ m each. How many pieces are obtained and how much is left? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A cake needs $\frac{3}{4}$ kg of flour. Rani wants to make only $\frac{2}{3}$ of the recipe. 5

(i) How much flour does she need?

(ii) How much flour is needed for 4 whole cakes?

(iii) How much flour is needed for $1\frac{1}{2}$ cakes?

(iv) She has 5 kg of flour. How many full cakes can she make and how much flour is left?

20. A farmer's land is 360 acres. He sells $\frac{2}{5}$ of it, gives $\frac{1}{6}$ of the remainder to his son and plants $\frac{4}{5}$ of what is left after that with sugarcane. Find each area and the area left unplanted. 5

OR

Pipe A fills a tank in 4 hours, pipe B in 12 hours, and a drain empties the full tank in 6 hours. Find the fraction filled in 1 hour with all three open, and the time to fill the empty tank. 5

END OF QUESTION PAPER — Total: 40 Marks